CpE 324

Methods of Research

Structuring Inquiries and Empirical Investigation...

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CONTENTS

01

Introducing Research

How the research process develops and how research is carried out...

01

Research Process

The means by which research is carried out

How the research process develops and how research is carried out...

The Research Process

01
Begin with an Idea. Refine that Idea.

02
State research question/statement in one sentence

03
Specific aim and objectives.

04
Literature Review

05
Select Methodology

06
Device data collection methods

07
Gather Data

08
Analyze Data

09
Draw Conclusions

10
Complete write-up of thesis/report

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Begin with an idea.

Refine that idea:
by thinking about it,
reflecting on it, and
by ***engaging with the literature***

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It is a good discipline to express your IDEA in ONE SENTENCE.

It becomes the CONCEPTUAL FRAMEWORK for your research project.

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Developing **an aim**,
and
a series of objectives.

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All published literature (theory) that you have read for your research.

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Research methodologies:
case study, survey,
attitude research, action research,
ethnographic research,
grounded theory research,
image-base research, and
feminist research

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Data Collection Methods:
observation, questionnaires,
interviews and focus groups,
scales, projective techniques,
diaries, images, narratives, life histories,
the use of documents, printed materials, film & symbols

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Data in research projects are
information **or evidence:**
to explore the phenomenon under investigation, or
to disprove or prove the hypothesis.

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> Analysis involves:
> First, by discribing the data
> then, interpreting it.

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CONCLUSIONS are drawn from the **analyzed data.**

These conclusions are then **theorized -- the body of knowledge grows & develops.**

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CONCLUSION is ESSENTIALLY a **JUDGEMENT**
or a **FINAL DECISION.**

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Complete write-up of thesis/report.

Finally, the write-up process is complete

United Nations' Sustainable Development Goals

DOST's HNRDA

2022-2028

H armonized
N ational
R esearch
D evelopment
A genda

DOST's HNRDA 2022-2028

Section1 National Integrated Basic Research Agenda (NIBRA) 2022-2028

A.Water Security - TUBIG Agenda (Tubig ay Buhayin at Ingatan)

B. Food & Nutition Security - SAPAT Agenda (Saganang Pagkain para sa Lahat)

C. Health Sufficiency - LIKAS Agenda (Likas Yaman para sa Kalusugan)

D. Clean Energy - ALERT Agenda (Alternatice Energy Research Trends)

1. Resource assessment of Alternative Energy (wind, solarm biofeuls, hydro)

E. Sustainable Communities - SAKLAW Agenda (Saklolo sa Lawa) Vulnerable Ecosystems(Lakes, rivers, wetlands, seas and oceans)

F. Inclusive Nation-Building - ATIN Agenda (Ang Tinig Natin)

2. Nat'l security & sovereignty

Cybersecurity (review of cybersecurity policies; IT and Communication; Data Protection in the government sector)

Section 2 Health Research and Development Agenda (2023-2028)

A. Tuklas Lunas (Drug Discovery & Development)

B. Functional Foods

C. Nutrition and Food Safety

D. Re-emerging and Emerging Diseases

E. Diagnostics

F. Omic Technologies for Health

G. Biomedical Devices Engineering for Health

DOST's HNRDA 2022-2028

Section 2 Health Research and Development Agenda (2023-2028)

H. Digital and Frontier Technologies for Health

I. Disaster Risk Reduction/Climate Change Adaptation in Health,

J. Mental Health

Section 3. Agriculture, Aquatic & Natural Resources (AANR) Research & Development Agenda - 2022 to 2028

A. Crops R&D Agenda -crop production system: smart farming approaches, development of climate-resilient technologies.

B. Livestock R&D Agenda -

C. Aquatic R&D Agenda

D. Forestry R&D Agenda

E. Natural Resources & Environment R&D Agenda - fish kill warning and mitigation system

F. Climate change adaptation & mitigation & disaster risk reduction - quick response S&T based assistance to AANR communities affected by hazards

G. Technology Transfer

H. Socio-economic, Policy & Governance

I. Human Development, Science Communication & Knowledge Mgmt

SMART FARMING S&T ROADMAP 2022-2032

SMART PROGRAM

Validation/Smartification of POT components for the economically important crops
Soilless agriculture
Drones used in surveillance system
Sensors and microcontrollers for smart irrigation techniques
Controlled environment agriculture
Trainings/capability building

Development of Smart Farming R and D Center

POT components:
Fertilizer and pest management system with automation

Sensors and microcontrollers for smart irrigation techniques

Irrigation system using AI to adopt to disaster risk issues
Sensors and automation for soilless technology

Technology Promotion
-Website Management
-News and Promotion
-School-On-Air
-Knowledge Management
-Field Days
-Trainings

Policy Research
Subsidized program for smart farming production

Standardization of the nutrient and structures , policies, data format, platforms, testing centers pf smart industry grade products
Patents and copyrights
Inclusion of smart farming in the early education curriculum

Institutional Partnerships
Public-private partnership
Organized farmers for smart farming production
Development of commercial models
Marketing

S&T Goal/s

- **To increase productivity, optimize product quality and profitability of selected crops using Smart Farming Technologies (SFT)**
- **To improve the capabilities of institutions in developing SFTs**
  To promote adoption of SFTs
- **To mainstream SFTs through enabling policy advocacies**
- **To establish knowledge management system for SFTs**
  To ensure Food Safety while using SFTs

2022 (P50)
2023 (P55)
2023 (P60)
2025 (P60)
2026 (P60)
2027 (P60)
2028 (P60)
2029 (P60)
2030 (P60)
2031 (P60)
2032 (P60)

Milestones
Programs on SF implemented by the government for productivity of selected crops on going but still lagging behind,
Some smart technologies using sensors, automation are available but high cost of investment
On going projects on Climate Smart Mechanization (automation) AIoT on Pest mgt, nutrient mgt.)

Milestones
Technology Pilot Tested (prototypes, systems)
Drones used in surveillance system
Drones used for spraying and fertilizer application
Controlled environment agriculture
Variable rate technology (fertilizer application, irrigation water)
AI /Data analytics

Milestones
Policy subsidy to adopt SF practices
private farms operated through smart farming technologies
Patents and market roadmap
- Upgraded fabrication and smart farming machine/ facilities

Milestones

Smart farming testing center established through joint partnership with privet sector
Impact studies conducted done

DOST's HNRDA 2022-2028

Section 4 Industry, Energy and Emerging Technology Research & Development Agenda 2022-2028

A. Additive Manufacturing
B. Advance Materials
C. Materials for Energy
D. Nanotechnology
E. Optics and Photonics
F. Electronics Industry
G. ICT Innovations

https://futurenetworks.ieee.org/

priorities for 2025-2026

1. Orbital angular momentum multiplexing (nanophotonics)
2. Radio over fiber upgrades
3. Satellite Communication Bands

priorities for 2027

1. Establishment of a 5G/6G ecosystem innovation center
2. Development of new technologies for use of higher bandwidths (ku-, K-, Ka-bands)

priorities for 2028

1. development of single core (nanocore) wireless mobile communication service

DOST's HNRDA 2022-2028

H. Industry 4.0 - requires convergence, interoperability, and connectivity of various emerging technologies including 3D printing, AI IoT, Cloud Computing, Augmented Reality, Blockchain, and many others.
It includes the concept of Smart City and society with people at its center. However the implementation of Industry 4.0 is difficult because of the numerous aspects that compose it. Not only must factories and cities adopt modern technologies, but people need to also be trained and prepare to use these technologies.

All these would require new and updated policies to handle cybersecurity, IP, and education issues. To address these issues, the plan for Industry 4.0 by focuses on three key elements:

connectivity, interoperability, and convergence.

priorities for 2024:

3. Integration of vertical and horizontal value chains
4. Development of the asset administration shell (ASS) as the interface of the digital and real machines
5. Development of generic cloud-based Manufacturing Execution System (MES) for smart manufacturing
6. Development of Supervisory, Control and Data Acquisition(SCADA) or automation system that can connect factory equipment.
7. Development of modules on SCADA to accelerate connectivity and automation
8. Develop digital transformation model for electronic industry companies

DOST's HNRDA 2022-2028

H. Industry 4.0

priorities for 2025:

1. Industry 4.0 demo lab
2. Asset Administration Shell (AAS) applications
3. Manufacturing Execution System (MES) for smart manufacturing
4. Supervisory, Control, and Data Acquisition(SCADA) or automation system
5. Digital transformation model

priorities for 2026:

1. Process visualization through augmented reality (AR)/Virtual reality(VR)
2. Development of cyber-physical production systems
3. Development of networked production & collaborative diagnostics and decision-making

priorities for 2027 to 2028:

1. Advanced production processes utilizing AR/VR
2. Industry 4.0 Architecture adoption for pilot factories in the regions
3. High-level Cyber-Physical Production Systems
4. Self-configuring, self-adjusting, self-optimizing systems
5. Intelligent applications AI for industry design Information Processing

DOST's HNRDA 2022-2028

I. Quantum Technology

J. Smart Cities - DOST aims to address challenges of urban and regional life in cities through the use of science, technology and innovation to enhance opportunities and addredd challenges to sustainable urban development against disasters.

TheDOST Smart and Sustainable Communities and Cities Framework aims to enhance research collaboration and to fund excelent research with lasting impact. The specific objective is to fully exploit the potential of the regions talent pool and maximize the benefits of an innovation-led economy with the following perspectives:

1. Integration of different dimensions of urban sustainability in the Framework of the UN Sustainable Development Goals.
2. Co-production - a way to extend research activities to bridge gaps between knowledge, understading, and action

priorities for 2023:

1. Creation of digital twin
2. Big Data, data mining, and data analytics
3. Service delivery optimization
4. Smart surveillance
5. Public sensor networks for situation monitoring
6. Multi-dimentional Data correlation

DOST’s
HNRDA
2022-2028

J. Smart Cities -
priorities for 2024:
1. Integrationd simulation and systhesis
2. Remote management
3. Internet of Things and Smart Systems
4. Collaborative diagnostics and decision-making
5. Smart grids
6. Decision intelligence
7. Prescriptive Analytics
8. Multi-dimensional data correlation
9. Earth observation(EO) solutions in public services

priorities for 2025:
1. Smart grids.
2. Decision intelligence
3. Prescriptive Analytics
4. Multi-dimensional data correlation
5. Earth observation(EO) solutions in public servicess

priorities for 2026:
2&3 of 2025 priorities
3. Recommender systems
4. FInancial securities recommendations
5. Intelligent Applications
6. Image & Video Recognition
7. Situation interpretation through heterogenous sensors

DOST's HNRDA 2022-2028

J. Artificial Intelligence

priorities for 2023:

1. Recommender systems
2. Financial securities recommendations intelligent applications
3. Image & Video Recognition
4. Situation interpretation through heterogenous sensors

priorities for 2024:

1. Reinforcement learning
2. Supervised, Unsupervised learning
3. Cognitive computing

priorities for 2025:

1. Cognitive computing - advanced intelligent applications such as virtual assistants and smart robots
2. Artificial General Intelligence

priorities for 2026:

1. Artificial Narrow Intelligence
2. Creative Machines
3. Diffractive Neural Networks
4. Technologies on data privacy and security
5. Cyber Defense Build-up

priorities for 2027 to 2028:

1. Artificial Super Intelligence
2. Simulation Engines
3. Swarm Artificial Intelligence
4. Quantum deep learning
5. Smart Data

DOST's HNRDA 2022-2028

L. CREATIVE INDUSTRIES

Creative Industries: game, animation and film

priorities for 2023:

1. Establishment pf “Pugad Sining”/Creative Innovation Hub for Graphics Design, Motion Capture, and Audio Post-Production
2. Development of serious games and gamification apps for tertiary education, and corporate sector
3. Development of proprietary software and softeare/platform-as-a-service
4. Development of algorithmic video editing
5. Automatic music generation and AI-assisted sound engineering

priorities for 2024:

1. Establishment of Extended Reality Laboratory
2. Industry-defined training on game theory and game design
3. Advancement in human-to-computer interfaces
4. Utilization of extended reality in mobile gaming applications

priorities for 2025:

1. Establishment of Interactive Moviemaking facility
2. Development of interactive moviemaking technology

priorities for 2026:

1. Capability and capacity building to produce hardware prototypes
2. Prototyping of advanced gaming devices and animation tools

DOST's
HNRDA
2022-2028

L. CREATIVE INDUSTRIES
Creative Industries: game, animation and film

priorities for 2027:
1.Prototyping of advanced hardware for animation and film development

priorities for 2028:
1. Intelligent screenplay writing
2. Development of an autonomous drone cinematography system
3. Development of policy on blockchain
4. Application of blockchain in the gaming industry
5. Establishment of Holographic Environment Simulator
6. Establishment of a creative city for game, animation, and film

M. SPACE TECHNOLOGY APPLICATIONS
1. Development of Earth Observation-based Smart City Decision Support Services

DOST’s HNRDA 2022-2028

N. TRANSPORTATION

Aim to alleviate traffic congestion, reduce accidents, reduce fuel consumption and vehicle emissions; and provide safer, efficient movement of people and goods and improve quality of life.

The transport R&D has 4 subsectors: Land, Maritime, Intelligent Transport Systems(ITS), and Logistics and Freight Management in order to address various specific problems per area concern.

Intelligent Transport Systems: By providing integrated and responsive transport systems using ICT, the Internet of Vehicles(IOV), cameras/sensors, an automatic fare collection system, and onboard trackers, it applies the use of computer programming and simulation software as decision-support tools.

a. Deployment of on-board IoT devices for PUV drivers
b. Design road incident detection software
c. Design integrated framework for transportation database connectivity
d. Launch the System Optimization and Routing Tool (SORT) app in a pilot city

DOST's HNRDA 2022-2028

O. CLEAN ENERGY GENERATION & STORAGE

1. Renewable generation
2. Renewable and sustainable chemistries, fuels, and feedstocks
3. Nuclear energy systems
4. Fusion Energy
5. Energy Efficiency & Conservation
6. Energy Storage

P. UTILITIES

1. Construction
2. Water Resources Management

DOST's
HNRDA
2022-2028

Q. DISASTER RISK REDUCTION-CLIMATE CHANGE ADAPTATION (DRR-CCA) pages 105 - 110

Lessening the impact and/or reducing the vulnerable of different communities to the harmful effects of natural hazrds (tsunami, earthquakes, volcanic activity, landslides, typhoons, thunderstorms, severe wind, heavy rains, and floods) and climate change (climate induced hazards - extreme weather phenomena[heatwaves, droughts, frost, hail, intense storms, etc.]; changes in precipatation and temperature patterns; and sea level rise). :

R. UNMANNED VEHICLE SYSTEMS (pages 110 -111
Unmanned aerial vehicles (UAV) are still among the latest technology trends sweeping across and disrupting a broad spectrum of industries. UVS has been the face of intelligence, disaster response and assessment, warfare/military, humanitarian relief, among others

S. FOOD pages 112 - 116
1. Food Safety Program
2. DOST Halal S&T Program
3. Food Innovation Program

T. METALS & ENGINEERING pages 116 - 118

Table of Contents

Undergraduate Thesis

TABLE OF CONTENTS

Undergraduate Thesis

continued...

CHAPTER II REVIEW OF RELATED LITERATURE

- Related Readings
- Related Literature
- Related Studies
- Synthesis & Justification
- Theoretical Framework
- Conceptual Framework

CHAPTER III METHODOLOGY

- Research Design
- Statistical Treatment
- IPO Model
- Prototype Development

REFERENCES

APPENDICES

Word Count

Undergraduate Thesis

Social Science Convention	Lay-out for a 20,000-word Thesis
Title Pages	Pages with name of the research, panel, dedication and appreciation
Abstract	Three hundred-word summary of the entire research, generally presented in one paragraph
Table of Contents	Including lists of tables, figures, photographs
Chapter I	Introduction, 2000 words A brief introduction to the entire thesis
Chapter II	Literature Review, 5000-words Contains the theoretical framework
Chapter III	Research Design & Methodology, 5000-words Contains the methodological framework
Chapter IV	Results & Discussion, 5000-words Contains the analytical framework
Chapter V	Conclusions & Recommendations, 3000-words Contains the fully developed, well-conceptualized conclusions from the research
References	Complete list using IEEE format
Appendices	Each appendix contains a document/table/figure referenced & discussed in the research

Test Of Researchability

01

The time needed to conduct the research:
To design the project,
Carry out the fieldwork,
Analyze the data,
Write up the findings,
Draw conclusions,
and Make recommendations.

02

The money needed to conduct the research:
Generally, funds are required to create your proto-type

03

The access to data:
Difficulties that researchers can encounter in accessing data, in attempting to access data, in securing access to data, and in maintaining access to data over time to complete the fieldwork

Introduction to Research Ethics

	Feb	March	April	May				
Focus on identifying research area. Develop idea in this area for research project. Turn this idea into a research question/statement								
Commence reading of literature. Develop aim and objectives. Decide on appropriate methodology for research, develop the data gathering methods to be used.								
Write research proposal. Secure permissions needed to conduct the research. Secure ethical approval the research.								
Write Chapter 1, Introduction (it is based on the research proposal). Begin writing literature review which is the Chapter 2								
Write methodology, Chapter 3								
Complete writing of the literature review. Finalize data gathering techniques (do this with reference to the literature)								
Title Defense								

Introduction to Research Ethics

	June	July	Aug	Sept	Oct	Nov	Dec.
Gather Data.							
Analyze Data.							
Write data analysis (Chapter IV).							
Write Conclusion & Recommendations (Chapter 5)							
Thesis Defense							
Complete the first draft, submit for feedback							
Complete second draft							
Final review of thesis, submit thesis							

THANK YOU